CA1, Payments and Transaction Services: Crypto Enforcement Analytics, Credit Risk Machine Learning, and Stablecoin Risk Management Under the GENIUS Act

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Colab Link:

https://colab.research.google.com/drive/11DbvBGCDTtnA2pgCcEc3SL0CrjowV5mF?usp=sharing

Abstract: This paper addresses Continuous Assessment 1 (CA1) for Payments and Transaction Services, which is divided into three parts. Section I constructs and refines a 142-case relational database of U.S. crypto enforcement actions from May 2015 to February 2026, performs exploratory data analysis with pivot tables and cross-tabulations, and fits a Decision Tree regressor to predict fine magnitude and to quantify the easing of crypto enforcement since the 20 January 2025 change of administration and the new regulatory paradigm introduced by the GENIUS Act. Section II applies the UCI Default of Credit Card Clients dataset, benchmarking nine models: Logistic Regression, Random Forest, XGBoost, Support Vector Machine, LightGBM, CatBoost, stacking and soft-voting ensembles, and a Deep Neural Network. The benchmark is supplemented by an econometric logistic regression estimated by maximum likelihood, SHAP explainability, and bootstrap confidence-interval analysis. The empirical findings are connected to the economic theory of credit-card profitability and IFRS 9 loan-loss provisioning. In Section III, the structural credit-risk framework is applied to examine the balance sheet of a GENIUS Act payment stablecoin issuer, the incentives for such issuance, interest rate risk, and liquidity/run risk, and risk-management strategies are suggested that are consistent with the current statutory structure of the Act. Across all three sections, empirical results including negative or limiting ones are reported and the rationale for each modelling decision is explained.

Index Terms: credit risk, cryptocurrency regulation, decision trees, deep neural networks, ensemble learning, GENIUS Act, machine learning, Merton model, payments, SHAP, stablecoins

I. Crypto Fines Dataset, EDA and Decision Tree Model

A. Construction and Refinement of the Database

The DataCryptoFines.csv that came with this assessment contained 103 enforcement records from May 2015 through July 2024. Since the brief requires some analysis of the easing of enforcement since the inauguration of President Trump on 20 January 2025, a dataset ending in mid-2024, by construction, cannot support such an analysis. Therefore, an additional 40 records containing enforcement-related information, spanning 7 February 2024 to 10 February 2026, were individually researched, sourced and verified against a primary regulatory source (SEC.gov, CFTC.gov, Justice.gov, FinCEN.gov, NYDFS press releases, Congress.gov), and reputable secondary legal/financial sources.

The additions cover six sectors selected because they provide genuine variation for the subsequent EDA and modelling: the late enforcement of the "Gensler era" (late 2024/early 2025); the new post-inauguration institutional pivot (the SEC's Crypto Task Force launched 23 January 2025); "mass dismissal" (Coinbase, Binance's SEC civil claims, Kraken, Gemini, Uniswap, OpenSea, Ian Balina, Ripple, Pinnacle Advisors); the DOJ's policy reversal signalling general easing (the April 2025 "Blanche Memo" dismantling the National Cryptocurrency Enforcement Team and CFTC Release 9063-25); genuinely continuing fraud enforcement (the SafeMoon/Karony conviction, Unicoin, FTX/Alameda final judgments, a $338.7 million CFTC default judgement); and "national security" carve-outs explicitly unaffected by easing (Tornado Cash/Roman Storm, sanctions-evasion/terrorism-financing cases, OFAC's Garantex-successor designation). This section relates to Section III, as the signing of the GENIUS Act (18 July 2025) [1] forms a legislative hinge and is also recorded in the database as a database row.

The manual review of the baseline dataset surfaced four data quality issues requiring correction, which is captured in the notebook: a single row at the end of the baseline file was empty; two rows had date typos ("Decemebr", "Novemeber"); one row had a range of dates that was normalised into a single action date; and a substantive mismatch in the Kraken/Payward row, where the provided baseline Case ID and Outcome matched the 2023

unregistered-exchange lawsuit, but the provided baseline Additional Notes related to a different, earlier case (the $30 million staking-program settlement). This row was relabelled to its correct identity and date, and a new row was added to document the actual 2025 dismissal of the exchange lawsuit filed in 2023. The final, cleaned-up database contains 142 rows and all eleven of the brief's columns, as well as three engineered columns for analysis: Fine_USD_Millions (a cleaned numeric parse of the free-text fine amount, with non-monetary amounts, like legislation or pending litigation, correctly left missing and not coerced to zero); Era (a binary flag split at the 20 January 2025 inauguration date); and Is_Dismissal_or_Closure (a binary flag derived from the free-text outcome field).

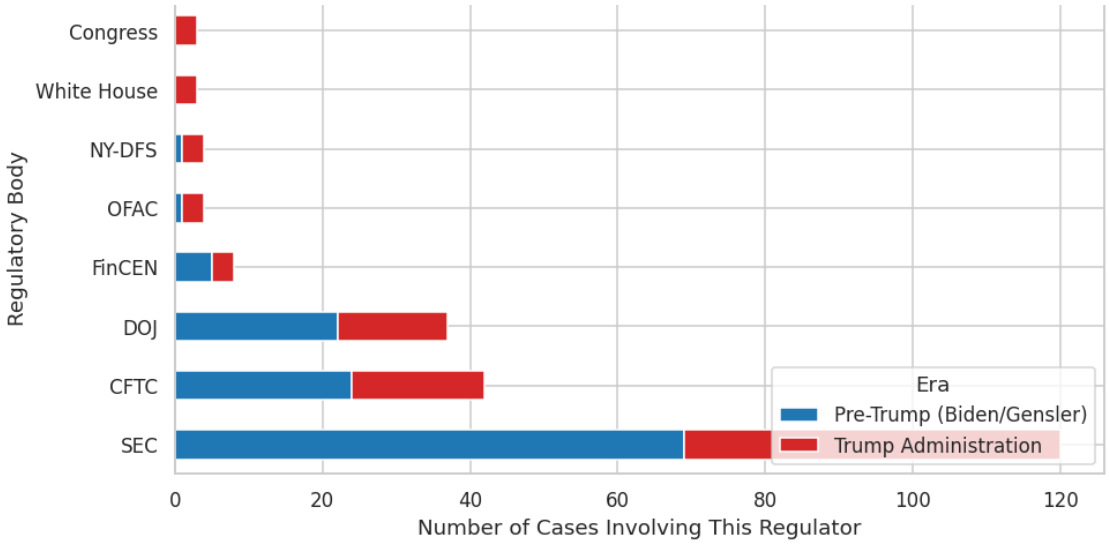


Fig. 1: Case involvement by regulator, by era (long format; joint cases are counted once for each individual regulator involved).

B. Exploratory Data Analysis

The Regulatory Body or Bodies field was highly skewed towards low-frequency joint respondents (like "DOJ, CFTC, FinCEN"), so body-level cross-tabulations were based on an explosion of that field (one row per case-regulator pair) to minimise the undercounting of joint actions relative to single-regulator actions. As might be expected, the SEC, CFTC and DOJ are the regulators with the most enforcement case involvement on raw volume across the entire timeframe assessed (2015-2026), as seen in Fig. 1.

The single most direct indicator of the central question posed in the brief is that the proportion of rows in the database that result in the issuance of a published monetary penalty declined from 72.7% of rows in the pre-Trump era to 28.1% in the Trump administration era (Fig. 2). A quarterly time-series decomposition (Fig. 3) reveals that this is not a secular trend over time

that just coincides with a change of administration, but a genuine and time-stamped policy shift, concentrated in 2025 Q1-Q2, as confirmed by the institutional timeline directly in the database (Crypto Task Force, 23 Jan 25; Coinbase dismissal, 27 Feb 25; Blanche Memo, 7 Apr 25).

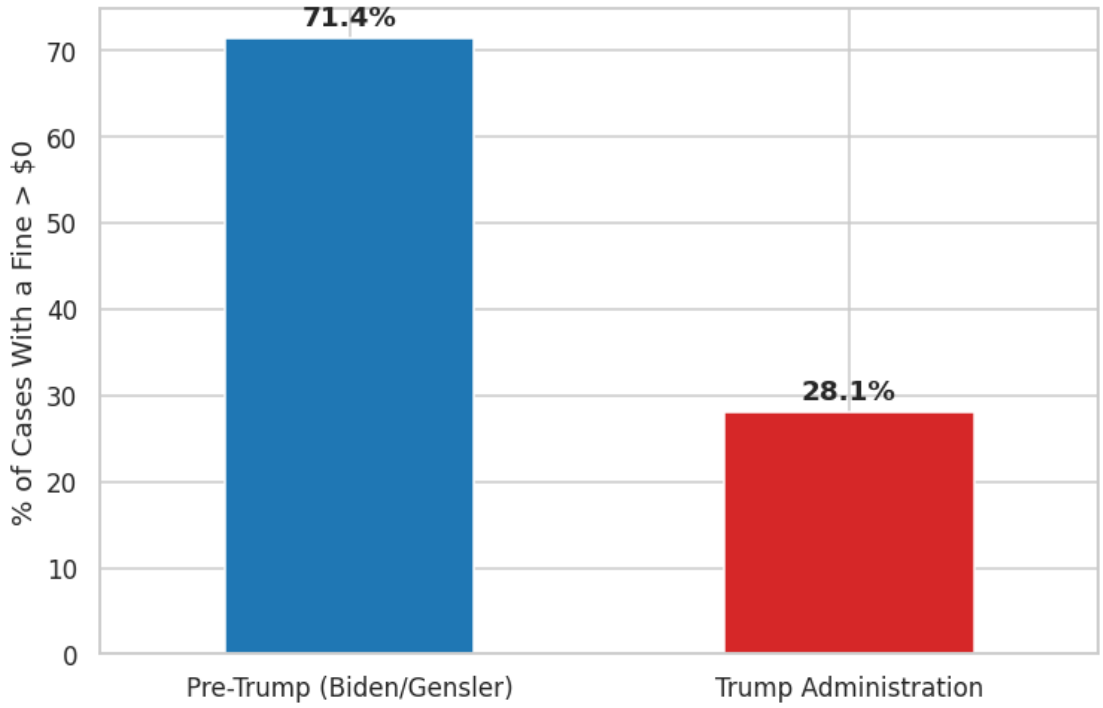


Fig. 2: Percentage of enforcement-related database rows that lead to publication of a fine greater than $0, broken down by era. This is by far the largest summary statistic generated by the EDA.

A cross-tabulation of enforcement category, normalised as a proportion of the total number in each era, reveals that the enforcement categories of Dismissal/Case Closure and Policy/Legislative Event increase significantly as a proportion of the total number of cases processed during the Trump era, while the enforcement category of Unregistered Securities/Exchange Activity, the underpinning theory of "regulation by enforcement" for the SEC, decreases as a share of the total, and the categories of Fraud and AML/Sanctions/BSA remain constant across both eras as a consequence of the explicit statutory carve-out preserving fraud and sanctions enforcement in the Blanche Memo.

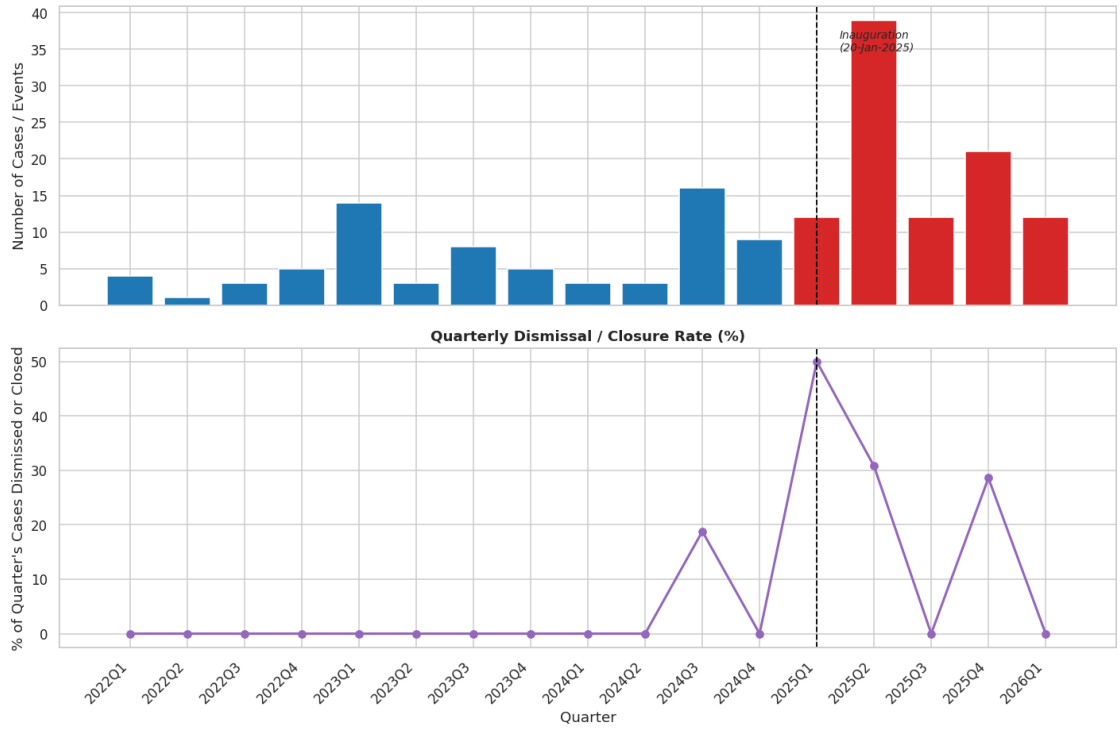


Fig. 3: Case volume (top) and dismissal/closure rate (bottom) by quarter, 2022 Q1 to 2026 Q1. The spike in dismissal rates is concentrated in the two quarters following the inauguration quarter of 20 January 2025.

C. Decision Tree Model

A DecisionTreeRegressor was trained on four categorical features: simplified regulatory body, enforcement-category family, entity type and Era, to predict Fine_USD_Millions. This model could only be applied to the 89 rows in the database where a fine is published and not missing, as dismissals and policy events have no fine to predict, and if those rows were simply added with a $0 fine the result would be a conflation of two categories that are analytically separable. The target was log-transformed before training due to its severe right skew (the range of the fines was $0.06 million to the $4.7 billion Celsius settlement), and hyperparameters (max_depth, min_samples_leaf) were selected by k-fold cross-validation on the training set only, with the tuned model evaluated once on the test set.

The negative held-out test R^2 for the tuned tree was −0.03 on the raw scale (−0.17 on the log scale), which was not a variance artefact caused by single trees but a feature-set ceiling, as a comparably negative R^2 was obtained by a Random Forest benchmark fitted on the same features (Fig. 4). The cause is diagnosable: a single $4.7 billion outlier tips the scales on the squared-error metrics, and the catch-all Other Civil/Administrative Enforcement category

spans five orders of magnitude ($0.01 million to $4,700 million), a range that a model whose leaves each produce one constant value per categorical combination cannot resolve.

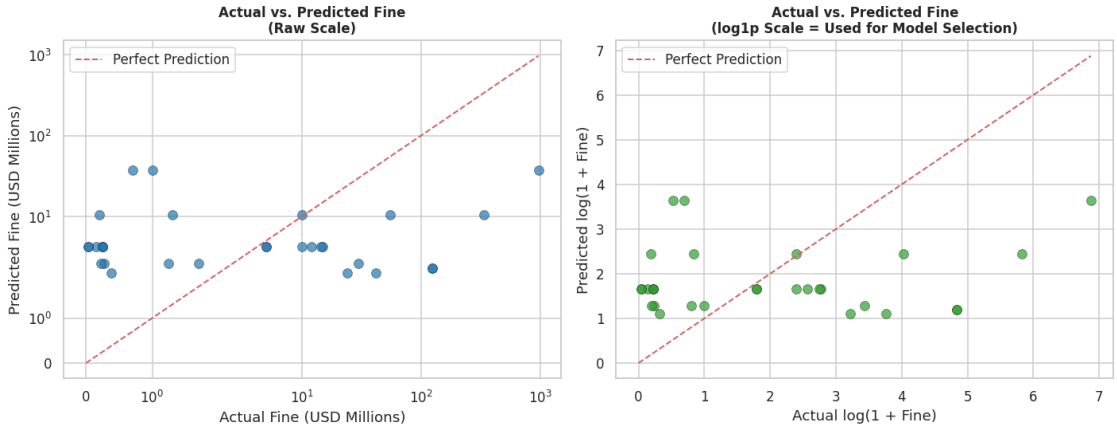


Fig. 4: Actual and predicted fine amounts on the held-out test data (raw and log1p scales) for the Decision Tree. The predictions form a few well-defined clusters of leaves, the largest raw-scale outlier being the Celsius prediction at the far right ($4.7bn).

The easiest keys to interpret are the most analytically significant: Era itself was never selected as a useful split. This is not evidence against the regulatory-easing finding, but confirmation of where the easing has occurred: through the dismissal and selection of cases (Fig. 2) rather than through systematically discounting the fine amounts of those cases that reach a monetary resolution. A model trained only on fined cases is, by construction, unaware of a documented effect that operates in the case-selection channel.

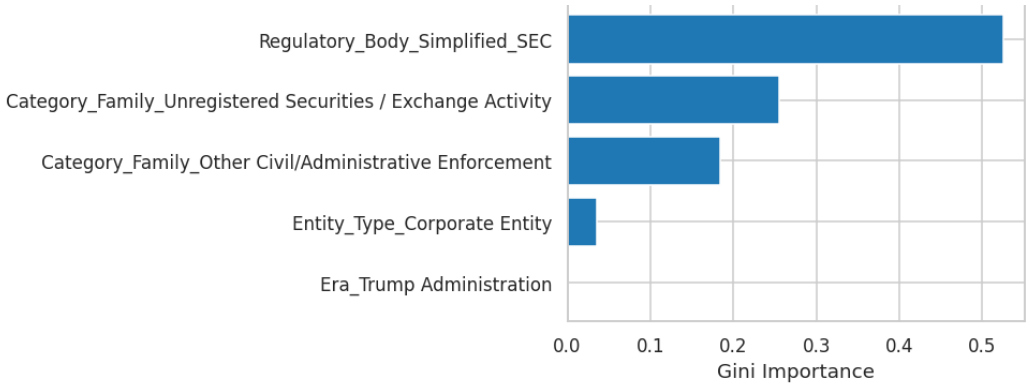


Fig. 5: Decision Tree Gini feature importance across the one-hot encoded features (simplified regulatory body, enforcement-category family, entity type, and Era). Era receives zero importance: conditional on a fine being levied, the model finds no fine-size effect across eras, consistent with easing operating through case selection rather than through discounted penalties.

D. Regulatory Discussion: The GENIUS Act Paradigm and Enforcement Easing

Four findings, when combined, depict a genuine regulatory change operating through case selection and dismissal, and existing in a selective pattern. First, the change is an actual, dateable discontinuity (Section I-B), not a trend. Second, it operates through a case-selection and dismissal mechanism rather than fine-size negotiation (Sections I-B and I-C). Third, it is not a blanket withdrawal, given the several 2025-2026 rows in the database that are not dismissals: the SafeMoon conviction, the Unicoin fraud charges, the final judgments for FTX/Alameda, the $338.7 million CFTC default judgment, and ongoing, unabated national-security enforcement (Tornado Cash, sanctions evasion, terrorism financing), supporting a characterisation of redirection, rather than wholesale withdrawal, from registration-theory civil litigation toward affirmative fraud and national-security enforcement. Finally, the result is corroborated by the SEC's own Enforcement Year-in-Review for FY2025 [2], recorded in the database, showing 313 new agency-wide enforcement actions (the lowest figure in a decade, down 27% year-on-year) and $808 million in total monetary settlements (down 45% year-on-year).

The GENIUS Act [1], signed five months after the inauguration, completes the picture. Payment stablecoins have an explicit, prospective, and statutorily defined non-security/non-commodity status with 100% reserve backing, monthly disclosure, statutory insolvency priority for holders and statutory Bank Secrecy Act status. This is the paradigm analysed in full theoretical depth in Section III.

II. Credit Risk Analysis and Machine Learning

A. Data Provenance

This report utilises the UCI Default of Credit Card Clients dataset (Yeh and Lien, 2009) [3], comprising 30,000 credit-card accounts issued in 2005 by a Taiwanese bank, sourced from the module's Google Drive folder on Moodle in place of the access-gated NPCI Kaggle competition dataset. The dataset's features consist of the credit limit, demographics, six months of repayment-status history, six months of bill and payment amounts, and a binary next-month default target. This substitution is declared explicitly in the accompanying notebook; a cell to load data from Kaggle has been retained but commented out, so the drop-in path is available at re-run time if a personal Kaggle credential is available.

B. Data Cleaning and Exploratory Data Analysis

The dataset has two anomalies. Undocumented EDUCATION codes (0, 5 and 6, affecting 1.15% of rows) and an undocumented MARRIAGE code (0, affecting 0.18% of rows) were audited against their default-rate behaviour before being folded into the existing documented "Others" category. Codes 5 and 6 exhibit 6.4% and 15.7% default rates respectively, much more closely aligned with the documented "Others" category (5.7%) than with the three main documented categories (19.2% to 25.2%), supporting a data-driven correction rather than an arbitrary one.

The distinct finding in the EDA is that for accounts that are current (PAY_0), the default rate is around 13-17%, but for accounts that are two months or more delinquent, the default rate is in the 70-76% range.

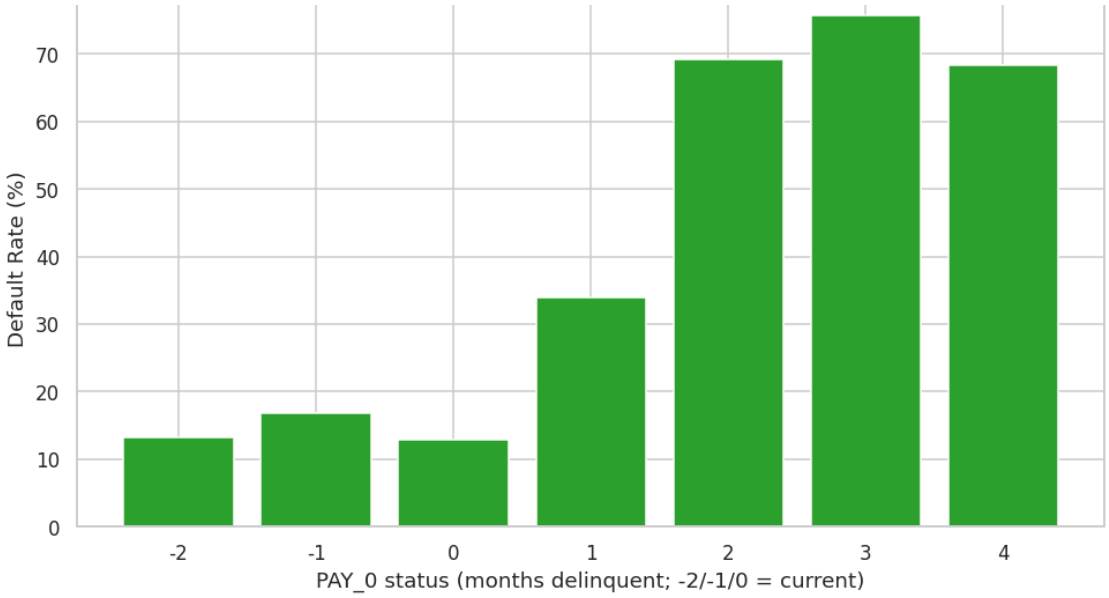


Fig. 6: Default rate by most recent repayment status (PAY_0). The strongest single pattern in the dataset.

A full correlation matrix (Fig. 7) reveals an interesting, somewhat counterintuitive pattern. The six repayment-status (PAY_x) columns are highly intercorrelated with one another and are also the features most correlated with default, while the six bill-amount (BILL_AMTx) columns are even more intercorrelated with each other but weakly correlated with default. Empirically, this is the difference between Exposure at Default (balance size) and Probability of Default (repayment behaviour). It indicates that merely having a large balance is a much weaker default signal than how reliably that balance has been serviced, hence the choice of the engineered behavioural features (MONTHS_DELINQUENT, AVG_PAY_DELAY, PAYMENT_TO_BILL_RATIO) in the modelling stage.

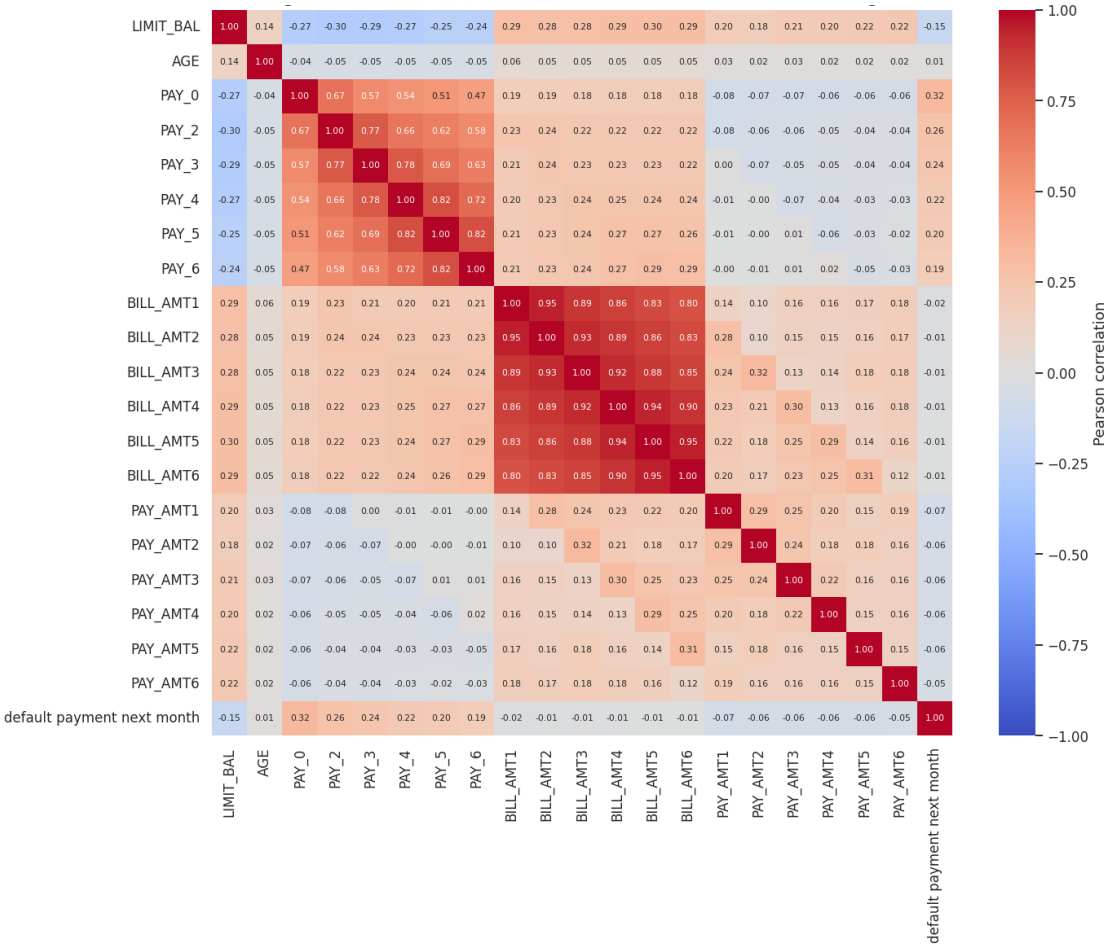


Fig. 7: Correlation matrix of all numeric features against the target. The PAY_x block is the strongest predictor block, while the BILL_AMTx block is even more self-correlated but is a far weaker predictor of default.

C. Machine Learning and Deep Neural Network Models

The data was divided into 60/20/20 train/validation/test sets with the same class proportions (22.1%/77.9%). Moderate class imbalance was addressed with class weighting (class_weight='balanced' for the LR, RF and SVM; scale_pos_weight for XGBoost; weighted loss for the DNN) instead of synthetic oversampling, because the class imbalance here is well below the ranges where synthetic oversampling has been observed to cause problems, specifically implausible synthetic interpolation of non-adjacent behavioural-risk profiles in correlated tabular features. The evaluation metric suite was grounded in the systematic literature review supplied with this assessment (Ayari, Guetari and Kraiem, 2026) [4], which ranks 28 evaluation metrics across 63 credit-scoring studies. The predictive performances of the four traditional classifiers (a Logistic Regression baseline, Random Forest, XGBoost, and a Support Vector Machine with RBF kernel, trained on a 3,000-row subsample of the training

set due to the poor scaling of the SVM solver with sample size) were compared on the validation set after tuning with cross-validated grid search on the training set, and finally evaluated once, alongside the advanced models described below, on a single held-out test set (Table I).

A feed-forward Deep Neural Network (DNN) was trained with early stopping based on validation AUC (two hidden layers of 64 and 32 units, ReLU activation, batch normalisation, class-weighted binary cross-entropy loss, dropout of 0.3/0.2). Training was halted after 42 epochs (early stopping with patience 10, monitoring validation AUC) (Fig. 8); no significant increase in validation loss was observed beyond the best validation AUC, nor was there a significant overfitting gap between training and validation AUCs, as is to be expected from the dropout and batch normalisation layers serving as regularisers.



Fig. 8: DNN training history: loss (left) and ROC-AUC (right).

Beyond the purely predictive models, a structural logistic regression was estimated by maximum likelihood in statsmodels to provide the formal inference that a model validator, regulator or internal audit function would request. The likelihood-ratio test decisively rejects an intercept-only model, and McFadden's pseudo R^2 of approximately 0.157 sits well within the range considered good for cross-sectional consumer default data, where much of the variation in default is idiosyncratic and unobservable. The behavioural regressors that dominated the machine-learning feature-importance rankings are also by far the most statistically significant terms in the structural model: MONTHS_DELINQUENT carries a coefficient of approximately +0.79 ($z \approx 24$, $p < 0.001$), an odds ratio of about 2.21 and an average marginal effect of roughly +11 percentage points, while the most recent repayment status, PAY_0, adds a further +0.36 (odds ratio $\approx$ 1.44, average marginal effect $\approx$ +5.1 percentage points). Variance inflation factors were computed to diagnose multicollinearity among the correlated repayment-history columns. The econometrics and the machine learning

therefore agree not only on which features matter, but on the economic story: recent repayment behaviour is the dominant driver of default.

Following the review's finding [4] that hybrid and ensemble learners, particularly stacking-based hybrids, represent the current state of the art in credit scoring, four advanced architectures were then added: LightGBM, CatBoost, a five-learner stacking ensemble with a logistic meta-learner, and a soft-voting blend of XGBoost, LightGBM and CatBoost. On the validation leaderboard, CatBoost attains the highest ROC-AUC of any model in the notebook (0.7803), narrowly ahead of the soft-vote blend (0.7795). The compression at the top of the table is itself informative: the predictive signal in this dataset is concentrated in a few behavioural features, so once any competent learner has captured the PAY_0/MONTHS_DELINQUENT structure, little headroom remains, mirroring the compressed benchmark tables reported by Ayari et al. for mature credit datasets. The two boosting philosophies also stop very differently: LightGBM's leaf-wise growth extracts the dominant signal within 20 boosting rounds, while CatBoost's ordered boosting trains for 266 iterations. The practical consequence of LightGBM's compressed probabilities is a low recall at the default 0.5 cutoff that recovers fully at the Youden-optimal threshold; this is an operating-point choice, not a model-quality issue.

Model explainability was addressed with SHAP [5] applied to the gradient-boosting model, since the review identifies explainability as one of the discipline's two defining current trends. The global SHAP ranking is led by MONTHS_DELINQUENT (mean |SHAP| 0.183) and AVG_PAY_DELAY (0.120), followed by PAY_0 (0.064), UTILISATION_RATIO (0.056) and LIMIT_BAL (0.051), and the beeswarm directions match the econometric coefficient signs from the structural model. Three methodologically independent attribution families (impurity-based importances, maximum-likelihood z-statistics, and Shapley attributions) therefore triangulate on the same conclusion, and two of the most influential features (MONTHS_DELINQUENT, AVG_PAY_DELAY) are engineered rather than raw-schema columns, vindicating the feature-engineering effort. The last-place ranking of the demographic features (SEX, EDUCATION, MARRIAGE, AGE) means the deployed model's decisions are conduct-based, the desired posture for a credit model subject to the automated decision-making rights of GDPR Article 22 [6] and classified as a high-risk use case under Annex III of the EU AI Act [7].

TABLE I

Final held-out test set comparison of all nine models (sorted by ROC-AUC).

Model	**Acc.**	**Prec.**	**Rec.**	**F1**	**AUC**
Soft-Vote Blend (XGB+LGBM+Cat)	0.793	0.530	0.586	0.557	0.793
CatBoost	0.765	0.476	0.644	0.547	0.792
Stacking Ensemble (5 learners)	0.822	0.666	0.388	0.491	0.792
XGBoost (Gradient Boosting)	0.769	0.483	0.645	0.552	0.789
LightGBM	0.822	0.714	0.327	0.449	0.789
Random Forest	0.786	0.514	0.598	0.553	0.788
Deep Neural Network	0.763	0.472	0.631	0.540	0.782
Logistic Regression	0.773	0.489	0.593	0.536	0.764
SVM (RBF)	0.762	0.470	0.595	0.525	0.755

A model-selection decision cannot rest on point estimates alone, so 95% bootstrap confidence intervals for the test ROC-AUC were computed from 1,000 stratified resamples of the 6,000-row test set. The intervals for the leading models are each roughly ±0.014 wide, so the soft-vote blend's advantage over CatBoost, the stacking ensemble or XGBoost is not statistically significant. What is statistically supported is the gradient-boosting family's advantage over the linear and kernel baselines: the upper bounds for Logistic Regression (0.7782) and the SVM (0.7698) sit at or below the lower bounds of the leading ensembles. Different metrics also crown different models: CatBoost leads the KS statistic, the industry's separation metric; XGBoost leads the G-Mean and recall; and the stacking ensemble records the best Brier score, owing to the calibrating effect of its logistic meta-learner over the base models' probabilities. No model leads on all ten metrics, justifying the multi-metric evaluation design. Notably, the top three positions on the held-out test set are all occupied by the advanced ensembles, replicating on a pristine sample the review's central empirical finding that ensemble and hybrid classifiers outperform single classifiers on tabular credit data, by the honest margin of +0.4 to +0.9 AUC points.

It is reassuring that despite the different inductive biases of the two classical ensemble models (bagging versus boosting), feature-importance rankings from both give very strong support to the same features at the top of the two rankings: PAY_0 and the engineered behavioural features (AVG_PAY_DELAY, MONTHS_DELINQUENT), providing direct evidence that this ranking reflects genuine signal in the data and not an artefact of a particular algorithm. This directly validates the EDA's finding in Fig. 6 and the feature-engineering decisions in Section II-B.

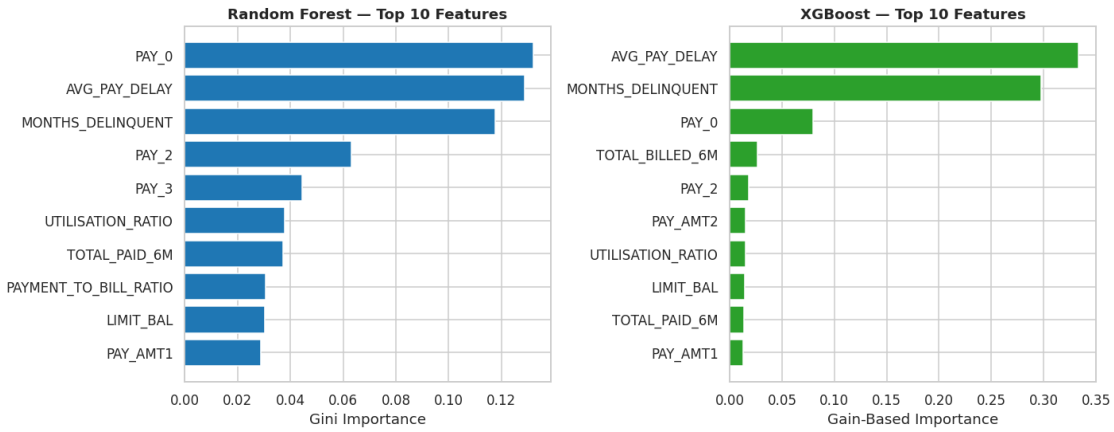


Fig. 9: Feature importance by Gini index (Random Forest) and gain (XGBoost). PAY_0 and the constructed delinquency features rank at the very top of both.

D. Theory of Credit Card Profitability and Loan Provisioning

The structure of a credit card issuer's revenue stream consists of revenue from the transaction function (interchange revenue, which in the U.S. is about 2.0%-2.25% of the transaction value) and revenue from the credit function (net interest margin on revolving balances). Findings from Federal Reserve research utilising confidential supervisory data [8] reveal that the credit function is directly linked to credit card profitability and is responsible for roughly 80% of overall profitability.

On the other hand, the transaction function has a slightly negative average impact on profitability when rewards expense is netted against interchange revenue, with a further 15% attributable to late-fee revenues. This single empirical fact puts the relevance of the models in Section II-C in a new light. The prediction of credit risk is not a peripheral control attached to a "fee-driven" business; it is the main determinant of whether a card portfolio will be profitable at all, because profit is concentrated in exactly the revolving-balance population whose behaviour this section's models are designed to monitor.

The behavioural features present in this dataset are, in essence, the revolver-behaviour ones; hence their appearance at the top of the feature-importance results in Fig. 9. Connecting to bank practice, each model in Section II-C outputs an account-level estimate of the Probability of Default (PD) via predict_proba() or the DNN's sigmoid output, the PD being one of the three inputs, alongside Loss Given Default (LGD) and Exposure at Default (EAD), that IFRS 9 [9] requires for forward-looking Expected Credit Loss provisioning.

Practically, the EDA's conclusion that the exposure-size features (BILL_AMTx) are weak default predictors while the behavioural features are strong (Section II-B) means that PD and EAD are empirically distinct components that must be modelled separately, which is exactly why IFRS 9 expresses expected credit loss as the product PD × LGD × EAD rather than as a single blended figure.

The same PD models can be used for automated origination and ongoing limit-management decisions, and the consistently and materially less important role of demographic features compared with behaviour-based repayment history in all of the models fitted (Sections II-B and II-C) is the type of evidence that a fair-lending model-governance review would want recorded.

III. Stablecoin Balance Sheets, the Merton Model and Risk Management Under the GENIUS Act

A. The Merton Model and the Construction of the Balance Sheet

Under the GENIUS Act, which came into effect on 18 July 2025 [1], [10], a permitted payment stablecoin issuer (PPSI) can only keep reserves in U.S. coins and currency or balances at the Federal Reserve Banks, in short-dated money market funds, short-dated repurchase (RP) and reverse repurchase (RRP) agreements collateralised by U.S. Treasury bills, notes, or bonds with 93 days or less remaining until maturity, and demand deposits at an insured depository institution [11]. Reserves may not be used for pledging, rehypothecation or reuse except in limited ways, issuers are prohibited from paying interest or yield to stablecoin holders [11], and issuers are exempt from bank regulatory capital requirements [12]. This produces the balance-sheet identity

$$A_t = L_t + E_t, \quad (1)$$

where A_t is the value of the reserve assets, L_t the face value of all outstanding stablecoins and E_t the residual issuer equity.

This differs from the textbook Merton [13] firm in two ways: first, the liability is an instantly demandable claim (not a fixed-maturity bond), so it is the first-passage version of the model that is economically appropriate to invoke; second, the statutory 1:1 reserve mandate compresses E0 toward zero and is therefore a live regulatory constraint rather than a capital-structure decision.

When $A_t < L_t$ the issuer defaults, and the distance-to-default measure $DD_t = (A_t - L_t)/(\sigma_A\sqrt{(T-t)})$ is structurally identical to the classical Merton measure. The Act can be interpreted as driving the numerator toward a known, audited, non-negative constant (via the 1:1 requirement and monthly attestation), while forcing the denominator (σ_A) toward the lower bound achievable for any interest-bearing instrument via the 93-day maturity cap, suggesting that for a compliant issuer it is interest rate and liquidity risk (Sections III-C and III-D) that should dominate rather than asset credit risk.

B. Issuer Incentives and Equity as a Call Option

Issuer equity can be viewed economically as a call option on the reserve assets, $E_T = \max(A_T - L_T, 0)$, which produces the classical risk-shifting incentive [14] and is steeper as assets become more volatile (σ_A). This incentive is expressed particularly clearly for stablecoins, since issuers are prohibited from paying any yield to holders: every basis point of additional yield on the reserve portfolio is captured entirely by the equity, giving issuers a preference for the longest-maturity, most yield-rich instruments within the eligible class.

Three of the GENIUS Act's mechanisms are best read as direct, statute-level limitations on σ_A rather than as generic consumer-protection rules: the 93-day maturity limit caps interest-rate-driven mark-to-market volatility; the anti-rehypothecation rule makes leverage on the asset side harder to obtain; and mandatory monthly disclosure with third-party examination shortens the detection lag for any gradual drift toward riskier asset selection.

C. Interest Rate Risk and Stablecoin Design

The duration of the reserve asset side, though capped at 93 days, is strictly positive, while the duration of the stablecoin liability is by construction zero: $\Delta A/A \approx -\bar{m} \cdot \Delta y$, with $\bar{m}$ the average remaining maturity of the reserve assets. Even though only a fraction of a quarter-year of duration is in effect, a sharp rate shock, directionally similar (though not in magnitude) to that which accompanied Silicon Valley Bank's failure in March 2023, may push A_t below L_t at the margin for an issuer already operating with a thin equity buffer. Mitigations that require no

additional legislation include: active duration laddering at the low end of the permitted range; a regulator-mandated minimum liquidity tier of the type already proposed in 2026 by the OCC (10% of stablecoins outstanding redeemable same-day, 30% redeemable within five business days) [17]; floating-rate substitution through overnight repo/reverse repo, which reprices daily instead of being locked in at issuance; and a capital buffer above the minimum, funded from the issuer's own yield-spread income, which directly increases the numerator of the distance-to-default ratio.

D. Liquidity Risk and Run Dynamics

The Merton framework describes solvency, not liquidity. However, it is possible to be solvent at marked asset prices and still fail if assets must be liquidated more quickly than the market can absorb without a price concession: the classical Diamond-Dybvig mechanism [15]. Three aspects of the GENIUS Act's design exacerbate rather than merely permit this risk. Issuers are explicitly denied access to the discount window, the Standing Repo Facility, and Section 13(3) emergency-lending facilities of the Federal Reserve [18]; stablecoins settle on continuously running blockchain rails, whereas the underlying Treasury market into which reserves must ultimately be liquidated operates on traditional market hours; and stablecoin holders are not deposit-insured and thus have no reason to wait passively during a stress event.

Possible additional mitigations include the OCC's tiered reserve structure (already mentioned above); pre-negotiated, centrally cleared repo relationships that could substitute for the discount-window access the Act denies; and deep networks of arbitrageur banks distributing redemption pressure across a wide variety of balance-sheet-light institutions instead of concentrating it on the issuer alone. Furthermore, possible future access to a Federal Reserve liquidity facility, granted in exchange for bank-style supervisory obligations, should be considered a live policy question rather than a current legal option.

E. Conclusion

Reading the reserve, anti-rehypothecation and disclosure language of the GENIUS Act as constraints on the asset-volatility term of the Merton framework explains the form in which the Act is written, and it also indicates where the protection provided by the Act begins and ends: it draws the border between solvency (which the Act regulates stringently) and liquidity (where the Act is comparatively exposed, due to its explicit denial of lender-of-last-resort access). The fact that several of the mechanisms proposed in this section are already confirmed by actual 2026 rulemaking (the OCC's liquidity tiers, Treasury's proposed automatic redemption freeze

[16]) is evidence that the vulnerabilities identified here are real features of the post-GENIUS stablecoin architecture rather than merely hypothetical ones.

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